

GED Math Practice Test 6

PART A — QUESTIONS

GED-MATH6-001

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Non-Calculator

Prompt:

A box contains 36 batteries. If $\frac{1}{6}$ of the batteries are rechargeable, how many batteries are rechargeable?

- A) 4
- B) 6
- C) 9
- D) 30

GED-MATH6-002

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Non-Calculator

Prompt:

What is the value of $3c^2 - 4$ when $c = 4$?

- A) 20
- B) 32
- C) 44
- D) 48

GED-MATH6-003

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Non-Calculator

Prompt:

Which fraction is equal to 0.875?

- A) $\frac{7}{8}$
- B) $\frac{3}{4}$
- C) $\frac{5}{8}$
- D) $\frac{1}{8}$

GED-MATH6-004

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Non-Calculator

Prompt:

Which expression is equivalent to $5(2x - 3) - 2(x + 4)$?

- A) $8x - 23$
- B) $8x - 7$

- C) $12x - 23$
D) $12x - 7$

GED-MATH6-005

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Non-Calculator

Prompt:

A tank holds 5 gallons of water. How many quarts of water does the tank hold?

- A) 10 quarts
B) 15 quarts
C) 20 quarts
D) 40 quarts

GED-MATH6-006

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

A science lab uses the formula $M = 12n - 7$ to estimate the mass M , in grams, after n samples are added to a tray. If $M = 77$, what is the value of n ?

- A) 5
B) 6
C) 7
D) 8

GED-MATH6-007

Section: Math

Type: numeric_entry

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

A rectangular banner has a length of 8.4 feet and a width of 2.5 feet. What is the area of the banner in square feet?

Type your answer as a decimal rounded to the nearest tenth.

GED-MATH6-008

Section: Math

Type: dropdown

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

The formula $R = 0.15B + 30$ gives a worker's total reimbursement R , in dollars, when B is the amount of eligible business expenses. In this formula, 0.15 represents [D1], and 30 represents [D2].

Dropdowns:

D1) the reimbursement rate for eligible expenses | the fixed reimbursement amount | the total reimbursement

D2) the reimbursement rate for eligible expenses | the fixed reimbursement amount | the eligible business expenses

GED-MATH6-009

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

A delivery route is 84 miles long. A driver completes $\frac{3}{7}$ of the route before lunch. How many miles are left after lunch?

- A) 24 miles
- B) 36 miles
- C) 48 miles
- D) 60 miles

GED-MATH6-010

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

Which value of x makes the inequality $4x + 9 < 37$ true?

- A) 5
- B) 7
- C) 8
- D) 10

GED-MATH6-011

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

The table shows the number of minutes a machine ran during four work periods.

Period: A, B, C, D

Minutes: 52, 61, 47, 68

What is the range of the minutes?

- A) 14
- B) 16
- C) 21
- D) 28

GED-MATH6-012

Section: Math

Type: multi_select

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

Select TWO answers.

Which expressions are equivalent to $12x - 18$?

- A) $6(2x - 3)$

- B) $3(4x - 6)$
- C) $2(6x - 18)$
- D) $12(x - 18)$
- E) $18 - 12x$

GED-MATH6-013

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

A box contains 6 small envelopes, 9 medium envelopes, and 15 large envelopes. If one envelope is chosen at random, what is the probability that it is not small?

- A) $1/5$
- B) $1/2$
- C) $4/5$
- D) $5/6$

GED-MATH6-014

Section: Math

Type: numeric_entry

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

The formula $A = 3b + 11$ gives the total number of art prints A after b batches are made. If $A = 83$, how many batches were made?

Type your answer as an integer.

GED-MATH6-015

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

A classroom floor is 28 feet long and 24 feet wide. The teacher wants to place tape along only three sides: both widths and one length. How many feet of tape are needed?

- A) 52 feet
- B) 76 feet
- C) 80 feet
- D) 104 feet

GED-MATH6-016

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

A table shows the value of a linear function.

x : 2, 5, 8, 11

y : 17, 29, 41, 53

What is the rate of change?

- A) 3
- B) 4
- C) 6
- D) 12

GED-MATH6-017

Section: Math

Type: multi_select

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

Select TWO answers.

A nutrition label lists 180 calories per serving and 7 grams of protein per serving. A person eats 2.5 servings. Which statements are true?

- A) The person eats 450 calories.
- B) The person eats 17.5 grams of protein.
- C) The person eats 72 calories.
- D) The person eats 14 grams of protein.
- E) The person eats 500 calories.

GED-MATH6-018

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

A number is divided by 5, and then 8 is added. The result is greater than 14. Which inequality represents this situation?

- A) $x/5 + 8 > 14$
- B) $5x + 8 > 14$
- C) $x/(5 + 8) > 14$
- D) $x/5 - 8 > 14$

GED-MATH6-019

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

A jacket is on sale for \$63 after a 30% discount. What was the original price?

- A) \$81.90
- B) \$90.00
- C) \$93.00
- D) \$210.00

GED-MATH6-020

Section: Math

Type: numeric_entry

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

Solve for y: $5y - 3(2y - 4) = 29$.

Type your answer as an integer.

GED-MATH6-021

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

A cylindrical container has a diameter of 10 inches and a height of 12 inches. Using $\pi \approx 3.14$, what is the volume of the container?

- A) 376.8 in^3
- B) 471.0 in^3
- C) 942.0 in^3
- D) $3,768.0 \text{ in}^3$

GED-MATH6-022

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

A plant's height is modeled by $H = 2.2w + 9.5$, where H is height in centimeters and w is the number of weeks after planting. What does 9.5 represent?

- A) The number of weeks observed
- B) The weekly growth rate
- C) The starting height of the plant
- D) The final height of the plant

GED-MATH6-023

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

A scale model of a bridge is 18 inches long. The actual bridge is 270 feet long. How many actual feet does 1 inch on the model represent?

- A) 12 feet
- B) 15 feet
- C) 18 feet
- D) 30 feet

GED-MATH6-024

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

Function A is represented by the equation $y = 5x - 4$. Function B is represented by the table.

x: 0, 2, 4, 6
y: 3, 11, 19, 27

Which statement is true?

- A) Function A has the greater rate of change.
- B) Function B has the greater rate of change.
- C) The two functions have the same rate of change.
- D) Function B has a negative rate of change.

GED-MATH6-025

Section: Math

Type: numeric_entry

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

A store sold a computer for \$742.50 after applying a 25% discount. What was the original price of the computer?

Type your answer as a decimal rounded to the nearest hundredth.

GED-MATH6-026

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

A community center sold 58 tickets for a workshop. Adult tickets cost \$14 and youth tickets cost \$9. The total ticket revenue was \$687. Which system can be used to find a , the number of adult tickets, and y , the number of youth tickets?

A) $a + y = 687$

$14a + 9y = 58$

B) $a + y = 58$

$14a + 9y = 687$

C) $14 + 9 = 58$

$a + y = 687$

D) $9a + 14y = 687$

$a + y = 687$

GED-MATH6-027

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

A rectangular lot is 45 meters long and 27 meters wide. A fence post is placed at each corner and every 9 meters along the perimeter. How many fence posts are needed?

A) 8

B) 16

C) 24

D) 32

GED-MATH6-028

Section: Math

Type: dropdown

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

The equation $3x + 2y = 18$ can be rewritten as $y = [D1]x + [D2]$.

Dropdowns:

D1) $-\frac{3}{2}$ | $\frac{3}{2}$ | $\frac{2}{3}$

D2) 6 | 9 | 18

GED-MATH6-029

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

A data set contains these values:

12, 18, 21, 23, 31

Which value must be added to make the mean of the six values equal to 22?

A) 17

B) 22

C) 27

D) 31

GED-MATH6-030

Section: Math

Type: multi_select

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

Select TWO answers.

Which statements are true about the equation $y = -1.5x + 12$?

A) When x increases by 2, y decreases by 3.

B) The y -intercept is 12.

C) The slope is 12.

D) When x increases by 1, y increases by 1.5.

E) The x -intercept is -1.5 .

GED-MATH6-031

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

A factory packs 1,728 screws equally into 36 boxes. Each box is then divided into 4 equal bags.

How many screws are in each bag?

A) 8

B) 12

- C) 24
- D) 48

GED-MATH6-032

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

Solve for n: $7(n + 2) - 4n = 50$.

- A) 8
- B) 10
- C) 12
- D) 14

GED-MATH6-033

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

A right triangle has one leg that is 20 centimeters long and a hypotenuse that is 29 centimeters long. What is the length of the other leg?

- A) 9 centimeters
- B) 18 centimeters
- C) 21 centimeters
- D) 25 centimeters

GED-MATH6-034

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

Solve the literal equation $4x - 3y = 12$ for y.

- A) $y = (4x - 12)/3$
- B) $y = (12 - 4x)/3$
- C) $y = 4x - 12$
- D) $y = 12 - 4x$

GED-MATH6-035

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

A water tank contains 1,200 gallons. It is draining at a constant rate. After 5 minutes, it contains 975 gallons. Which equation models the amount of water W left after t minutes?

- A) $W = 1,200 - 45t$
- B) $W = 1,200 - 225t$

- C) $W = 975 - 45t$
D) $W = 45t + 975$

GED-MATH6-036

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

A contractor mixes concrete using 2 bags of cement for every 5 bags of gravel. If the contractor uses 42 bags of cement, how many bags of gravel are needed?

- A) 84
B) 95
C) 105
D) 147

GED-MATH6-037

Section: Math

Type: drag_drop

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

Match each target with the correct tile.

Tiles:

- A) $y = 2(x + 5)$
B) $y = 2x + 5$
C) $y = 5x + 2$
D) $y = 5(x + 2)$

Targets:

- T1) Add 5 to a number, then multiply the result by 2
T2) Multiply a number by 2, then add 5
T3) Add 2 to a number, then multiply the result by 5

GED-MATH6-038

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

A medication dosage is 2.4 milligrams per kilogram of body weight. A patient weighs 75 kilograms. How many milligrams of medication are needed?

- A) 31.25 mg
B) 72.6 mg
C) 180 mg
D) 312.5 mg

GED-MATH6-039

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

Which ordered pair satisfies the inequality $y > 2x + 3$?

- A) (1, 4)
- B) (2, 6)
- C) (3, 10)
- D) (4, 11)

GED-MATH6-040

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

A quadratic function is defined by $p(x) = (x - 3)^2 + 4$. What is $p(8)$?

- A) 21
- B) 25
- C) 29
- D) 53

GED-MATH6-041

Section: Math

Type: multi_select

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

Select TWO answers.

A table shows the number of miles driven and gallons of fuel used by a truck.

Miles driven: 120, 240, 360

Gallons used: 8, 16, 24

Which statements are true?

- A) The truck uses 1 gallon for every 15 miles.
- B) The truck travels 15 miles per gallon.
- C) The truck travels 12 miles per gallon.
- D) The truck uses 8 gallons for every 240 miles.
- E) The relationship is not proportional.

GED-MATH6-042

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

For the function $f(x) = -2x^2 + 7x + 6$, what is $f(-1)$?

- A) -3
- B) 1
- C) 11
- D) 15

GED-MATH6-043

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

Which table represents a linear relationship with a rate of change of 0.75?

A) x: 0, 4, 8, 12

y: 2, 5, 8, 11

B) x: 0, 4, 8, 12

y: 3, 6, 10, 15

C) x: 0, 4, 8, 12

y: 1, 4, 6, 7

D) x: 0, 4, 8, 12

y: 0.75, 1.5, 3, 6

GED-MATH6-044

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

A population of insects is modeled by $N = 500(0.8)^t$, where t is the number of weeks. What does 0.8 mean in this model?

A) The population decreases by 20% each week.

B) The population increases by 80% each week.

C) The starting population is 80 insects.

D) The population decreases by 80 insects each week.

GED-MATH6-045

Section: Math

Type: numeric_entry

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

The formula $S = 2\pi rh + 2\pi r^2$ gives the surface area S of a cylinder. Using $\pi \approx 3.14$, what is S when $r = 3$ and $h = 10$?

Type your answer as a decimal rounded to the nearest tenth.

GED-MATH6-046

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

A city budget increased from \$2.4 million to \$2.7 million. What was the percent increase?

A) 10%

B) 12.5%

C) 15%

D) 30%

PART B — ANSWER KEY + EXPLANATIONS

GED-MATH6-001 — Correct: B — Correct Answer: 6 — Explanation: One sixth of 36 is $36 \div 6 = 6$.

GED-MATH6-002 — Correct: C — Correct Answer: 44 — Explanation: Substitute $c = 4$: $3(4^2) - 4 = 3(16) - 4 = 44$.

GED-MATH6-003 — Correct: A — Correct Answer: $7/8$ — Explanation: 0.875 equals $875/1000$, which simplifies to $7/8$.

GED-MATH6-004 — Correct: A — Correct Answer: $8x - 23$ — Explanation: Distribute and combine like terms: $5(2x - 3) - 2(x + 4) = 10x - 15 - 2x - 8 = 8x - 23$.

GED-MATH6-005 — Correct: C — Correct Answer: 20 quarts — Explanation: There are 4 quarts in 1 gallon, so 5 gallons = 20 quarts.

GED-MATH6-006 — Correct: C — Correct Answer: 7 — Explanation: Substitute $M = 77$: $77 = 12n - 7$. Add 7 to get $84 = 12n$, so $n = 7$.

GED-MATH6-007 — Correct: 21.0 — Tolerance: exact — Correct Answer: 21.0 — Explanation: Area = $8.4 \times 2.5 = 21.0$ square feet.

GED-MATH6-008 — Correct: D1=the reimbursement rate for eligible expenses; D2=the fixed reimbursement amount — Correct Answer: D1=the reimbursement rate for eligible expenses; D2=the fixed reimbursement amount — Explanation: The coefficient 0.15 is the reimbursement rate, and the constant 30 is the fixed amount.

GED-MATH6-009 — Correct: C — Correct Answer: 48 miles — Explanation: The driver completes $3/7$ of 84, which is 36 miles. The miles left are $84 - 36 = 48$.

GED-MATH6-010 — Correct: A — Correct Answer: 5 — Explanation: Solve $4x + 9 < 37$ to get $4x < 28$, so $x < 7$. Only 5 makes the inequality true.

GED-MATH6-011 — Correct: C — Correct Answer: 21 — Explanation: The range is $68 - 47 = 21$.

GED-MATH6-012 — Correct: A,B — Correct Answer: $6(2x - 3) \mid 3(4x - 6)$ — Explanation: Both expressions distribute to $12x - 18$.

GED-MATH6-013 — Correct: C — Correct Answer: $4/5$ — Explanation: There are 30 envelopes total, and 24 are not small. The probability is $24/30 = 4/5$.

GED-MATH6-014 — Correct: 24 — Tolerance: exact — Correct Answer: 24 — Explanation: Substitute $A = 83$: $83 = 3b + 11$. Subtract 11 to get $72 = 3b$, so $b = 24$.

GED-MATH6-015 — Correct: B — Correct Answer: 76 feet — Explanation: Tape is needed for two widths and one length: $24 + 24 + 28 = 76$ feet.

GED-MATH6-016 — Correct: B — Correct Answer: 4 — Explanation: When x increases by 3, y increases by 12, so the rate of change is $12 \div 3 = 4$.

GED-MATH6-017 — Correct: A,B — Correct Answer: The person eats 450 calories. | The person eats 17.5 grams of protein. — Explanation: Calories: $180 \times 2.5 = 450$. Protein: $7 \times 2.5 = 17.5$ grams.

GED-MATH6-018 — Correct: A — Correct Answer: $x/5 + 8 > 14$ — Explanation: “A number is divided by 5” is $x/5$, and then 8 is added.

GED-MATH6-019 — Correct: B — Correct Answer: \$90.00 — Explanation: After a 30% discount, the sale price is 70% of the original price. So $63 \div 0.70 = 90$.

GED-MATH6-020 — Correct: -17 — Tolerance: exact — Correct Answer: -17 — Explanation: $5y - 3(2y - 4) = 5y - 6y + 12 = -y + 12$. Then $-y + 12 = 29$, so $-y = 17$ and $y = -17$.

GED-MATH6-021 — Correct: C — Correct Answer: 942.0 in^3 — Explanation: The radius is 5 inches. Volume = $\pi r^2 h = 3.14(5^2)(12) = 942.0$ cubic inches.

GED-MATH6-022 — Correct: C — Correct Answer: The starting height of the plant — Explanation: In $H = 2.2w + 9.5$, 9.5 is the height when $w = 0$.

GED-MATH6-023 — Correct: B — Correct Answer: 15 feet — Explanation: Divide the actual length by the model length: $270 \div 18 = 15$ feet per inch.

GED-MATH6-024 — Correct: A — Correct Answer: Function A has the greater rate of change. — Explanation: Function A has rate 5. Function B has rate $(11 - 3)/(2 - 0) = 4$, so Function A has the greater rate.

GED-MATH6-025 — Correct: 990.00 — Tolerance: exact — Correct Answer: 990.00 — Explanation: The sale price is 75% of the original price, so $742.50 \div 0.75 = 990.00$.

GED-MATH6-026 — Correct: B — Correct Answer: $a + y = 58$
 $14a + 9y = 687$ — Explanation: The first equation represents the number of tickets, and the second represents the total revenue.

GED-MATH6-027 — Correct: B — Correct Answer: 16 — Explanation: The perimeter is $2(45 + 27) = 144$ meters. A post every 9 meters around a closed perimeter gives $144 \div 9 = 16$ posts.

GED-MATH6-028 — Correct: $D1=-3/2$; $D2=9$ — Correct Answer: $D1=-3/2$; $D2=9$ — Explanation: $3x + 2y = 18$ gives $2y = -3x + 18$, so $y = -3/2x + 9$.

GED-MATH6-029 — Correct: C — Correct Answer: 27 — Explanation: Six values with mean 22 must total 132. The five given values total 105, so the added value is $132 - 105 = 27$.

GED-MATH6-030 — Correct: A,B — Correct Answer: When x increases by 2, y decreases by 3. | The y -intercept is 12. — Explanation: The slope -1.5 means y decreases by 1.5 for each increase of 1 in x , so an increase of 2 in x decreases y by 3. The y -intercept is 12.

GED-MATH6-031 — Correct: B — Correct Answer: 12 — Explanation: Each box has $1,728 \div 36 = 48$ screws. Each bag has $48 \div 4 = 12$ screws.

GED-MATH6-032 — Correct: C — Correct Answer: 12 — Explanation: Distribute: $7n + 14 - 4n = 50$, so $3n + 14 = 50$. Then $3n = 36$ and $n = 12$.

GED-MATH6-033 — Correct: C — Correct Answer: 21 centimeters — Explanation: The missing leg is $\sqrt{(29^2 - 20^2)} = \sqrt{(841 - 400)} = \sqrt{441} = 21$.

GED-MATH6-034 — Correct: A — Correct Answer: $y = (4x - 12)/3$ — Explanation: From $4x - 3y = 12$, subtract $4x$ to get $-3y = 12 - 4x$. Dividing by -3 gives $y = (4x - 12)/3$.

GED-MATH6-035 — Correct: A — Correct Answer: $W = 1,200 - 45t$ — Explanation: The tank loses $1,200 - 975 = 225$ gallons in 5 minutes, or 45 gallons per minute.

GED-MATH6-036 — Correct: C — Correct Answer: 105 — Explanation: The ratio cement to gravel is 2:5. Since 42 is 21 times 2, the gravel amount is $21 \times 5 = 105$ bags.

GED-MATH6-037 — Correct: T1=A; T2=B; T3=D — Correct Answer: T1= $y = 2(x + 5)$; T2= $y = 2x + 5$; T3= $y = 5(x + 2)$ — Explanation: Parentheses show which operation happens before multiplication.

GED-MATH6-038 — Correct: C — Correct Answer: 180 mg — Explanation: Multiply 2.4 mg/kg by 75 kg: $2.4 \times 75 = 180$ mg.

GED-MATH6-039 — Correct: C — Correct Answer: (3, 10) — Explanation: Substitute $x = 3$: $2(3) + 3 = 9$, and $10 > 9$.

GED-MATH6-040 — Correct: C — Correct Answer: 29 — Explanation: $p(8) = (8 - 3)^2 + 4 = 5^2 + 4 = 29$.

GED-MATH6-041 — Correct: A,B — Correct Answer: The truck uses 1 gallon for every 15 miles. | The truck travels 15 miles per gallon. — Explanation: For each pair, miles divided by gallons equals 15, so both statements describe the same unit rate.

GED-MATH6-042 — Correct: A — Correct Answer: -3 — Explanation: $f(-1) = -2(-1)^2 + 7(-1) + 6 = -2 - 7 + 6 = -3$.

GED-MATH6-043 — Correct: A — Correct Answer: x: 0, 4, 8, 12
y: 2, 5, 8, 11 — Explanation: In choice A, y increases by 3 when x increases by 4, so the rate is $3 \div 4 = 0.75$.

GED-MATH6-044 — Correct: A — Correct Answer: The population decreases by 20% each week. — Explanation: Multiplying by 0.8 means 80% of the population remains each week, so the decrease is 20%.

GED-MATH6-045 — Correct: 244.9 — Tolerance: ± 0.1 — Correct Answer: 244.9 — Explanation: $S = 2\pi rh + 2\pi r^2 = 2(3.14)(3)(10) + 2(3.14)(3^2) = 188.4 + 56.52 = 244.92$, which rounds to 244.9.

GED-MATH6-046 — Correct: B — Correct Answer: 12.5% — Explanation: The increase is $2.7 - 2.4 = 0.3$ million. Then $0.3 \div 2.4 = 0.125$, or 12.5%.